MIL-M-10452C
3 July 1984
SUPERSEDING
MIL-M-10452B
30 March 1977

MILITARY SPECIFICATION

MONOCHLOROACETIC ACID, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one technical grade of monochloroacetic acid.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-D-723	-	Drums, Fiber
PPP-P-704	-	Pails, Metal: (Shipping, Steel, 1 Through 12 Gallons)

MILITARY

MIL-B-117	-	Bags, Sleeves and Tubing - Interior Packaging

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1168 - Ammunition Lot Numbering

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publication should cite "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

D1193 - Reagent Water
E1 - ASTM Thermometers

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Appearance. Monochloroacetic acid shall be crystalline and colorless to light-brown when tested as specified in 4.2.4.1.

3.2 Chemical and physical characteristics. Monochloroacetic acid shall conform to the chemical and physical characteristics of table I when tested as specified therein.

TABLE I. Chemical and physical characteristics

Characteristic	Requirement		Test paragraph
	Minimum	Maximum	
Freezing point, °C	60.0	---	4.2.4.2
Monochloroacetic acid, percent by weight	97.0	---	4.2.4.3
Dichloroacetic acid, percent by weight	---	1.5	4.2.4.3
Acetic acid, percent by weight	---	1.5	4.2.4.3

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the monochloroacetic acid produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled be in accordance with MIL-STD-1168.

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4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For test. See 6.5 for sampling and testing precautions. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 200 grams (g) shall be removed from each sample container and placed in a clean, dry container labeled to identify the lot and container from which it was taken.

TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit container, ready for shipment. Sample unit containers shall be examined for the following defects using an AQL of 1.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Marking incorrect, missing, or illegible
(f) Unitization not as specified

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. See 6.5 for sampling and testing precautions. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

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4.2.4.1 Appearance. Visually examine the specimen for form and color.

4.2.4.2 Freezing point. Fill half of an oven-dried, 25-millimeter (mm) by 200-mm ground glass-stoppered test tube with specimen and stopper. Immerse approximately 5 inches of the tube in a water bath maintained at 62.5° to 63.0°C until the specimen is melted. This may require as long as 24 hours. Allow the tube to remain in the bath until the specimen in thoroughly melted. Remove the tube from the bath and insert it into a larger tube. Place fiberglass or other suitable material between the two tubes to prevent contact between them. Remove the stopper from the tube containing the melted specimen and immediately insert a thermometer conforming to No. 36C of ASTM E1 to the required depth. Stir the melted specimen with the thermometer. (The melted specimen will first supercool to approximately 58°C.) When crystallization starts, the temperature will rise to a maximum which will persist long enough for a reading to be taken. Record this temperature and report it as the freezing point.

4.2.4.3 Monochloroacetic acid, dichloroacetic acid, and acetic acid.

4.2.4.2.1 Apparatus. The chromatographic apparatus used for this procedure consists of a glass tube 14 mm inside diameter and 50 centimeters long, provided with a constriction at one end and a B 19 socket at the other. The solvent reservoir is provided with a B 19 joint to mate with the column joint. Titrations are carried out using a 10-milliliter (mL) buret calibrated in 0.02-mL increments. Estimations may be made to 0.01 mL which is the size of one drop of titrant. A fine glass jet serves to deliver a stream of CO_2 free nitrogen under the solution titrated.

4.2.4.3.2 Reagents. All reagents shall be analytical reagent grade unless otherwise specified.

(a) Isopropyl ether - technical grade
(b) Petroleum ether - boiling range 60° to 80°C
(c) Sodium hydroxide solution - approximately 1N
(d) Sulfuric acid - 1N
(e) Sodium methylate - 0.01N: Weigh about 0.6 g of bright metallic sodium, wash rapidly with two 10-mL portions of methanol. Then allow to dissolve in 100 mL of methanol. When dissolved, dilute with an additional 150 mL of methanol and add this mixture to 1,500 mL of benzene through which a stream of nitrogen has been passed for between 1/4 to 1/2 hour to remove carbon dioxide. Store in an aspirator protected by a soda-asbestos guard tube. Standardize against pure monochloracetic acid.
(f) Thymol blue indicator - 1 g per liter in ethanol neutralized periodically with sodium methylate solution.
(g) Equilibrated solvent (S) - Mix equal volumes of isopropyl ether and petroleum ether and extract about 2 liters of the mixture with a two 100-mL portions of sodium hydroxide solution. Discard the aqueous layers and wash with a 100-mL portion of water. Discard the water wash and equilibrate the solvent with 50 mL of 1N sulfuric acid.

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(h) Supporting medium - Mix "Hyflo super-cel" (see 6.6) intimately with an approximately equal weight of 1N sulfuric acid. Reflux the mixture with a large excess of ether for 30 to 60 minutes. Cool and filter rapidly on a large Buchner funnel. Wash with copious amounts of water until the washings are neutral to bromocresol green indicator. Suck as dry as possible on the pump and dry the Hyflo on a flat dish at 105° to 110°C for 24 hours. Stir periodically with a spatula to dry the Hyflo efficently. Cool to room temperature.

4.2.4.3.3 Preparation of the column. Weigh 20 g of prepared Hyflo super-cel into a mortar. Add dropwise from a pipet with stirring, 10 mL of 1N sulfuric acid. Triturate gently but thoroughly to homogenize the Hyflo-sulfuric acid mixture. Add sufficient equilibrated solvent to form a thin slurry and mix well. Having plugged the bottom of the column with a small wad of ether-extracted cotton wool, pack the slurry into the column using Martin's method (see 6.7) with a perforated disc plunger. The length of a normal column is 24 to 28 cm. Wash through with about 50 mL of solvent S using an applied pressure, at 2.5 to 3 mL per minute.

4.2.4.3.4 Procedure.

(a) Standardization. Prepare a solution (A) of acids in isopropyl ether-petroleum ether containing approximately 20 g monochloroacetic acid, 2 g dichloroacetic acid, and 5 g acetic acid per liter. By applying nitrogen pressure to the top of the column, force the solvent carefully through the column until the surface of the solvent just reaches the surface of the packing. Carefully release the pressure and pipet 1 mL of solution A on to the surface. Gently apply pressure again to force the 1 mL into the packing. Release the pressure and wash the solution onto the column with two 1-mL portions of equilibrated solvent S. Pass equilibrated solvent through the column at a rate of 2.5 to 3.0 mL per minute by application of pressure to the top of the solvent reservoir. Collect 5-mL fractions in test tubes and titrate each fraction with 0.01N sodium methylate under a stream of nitrogen. When the dichloroacetic and monochloroacetic acids have been eluted, collect 20-mL fractions and titrate in the same way until a constant blank value (usually 0.02 to 0.03 mL) is obtained. The peak for dichloroacetic acid occurs at approximately 25 mL. The peak for monochloroacetic acid occurs at approximately 50 mL. The peak for acetic acid occurs at approximately 150 mL. The position of the peaks gives an indication of where fractions should be collected.

(b) Determination of unknown sample. Dissolve about 2 g (W) of commercial monochloroacetic acid in equilibrated solvent and make up to 100 mL in a standard flask. Pipet 1 mL of this solution on to the column and wash on with two 1-mL portions of solvents. Elute with solvent S at 2.5 to 3.0 mL per minute. Collect and titrate fractions as specified in 4.2.4.3.4(a). When experience has been gained, larger fractions can be taken with ease in the following suggested manner. Collect 5-mL fractions until the two dichloroacetic acid fractions are titrated (after 20 mL). Following the next blank

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fraction, collect a suitable fraction (usually about 40 mL) indicated by the standardization to contain all the monochloroacetic acid. Check the next 5-mL fraction as a blank and then collect 20-mL fractions until the acetic acid fraction is expected. Collect sufficient further solvent indicated by the standardization to contain all the acetic acid and titrate with sodium methylate. Check the next 5-mL fraction as a blank and then a fraction of the same volumne as the acetic acid fraction. This should be used as the blank for the acetic acid fraction. The blank for monochloroacetic acid is obtained similarly by titrating the appropriate volume of solvent.

4.2.4.3.5 Calculation. Calculate the percent by weight dichloroacetic acid, monochloroacetic acid, and acetic acid as follows:

$$\text{Percent dichloroacetic acid} = \frac{12.9\ V}{W}$$

where: V = The sum of the two first large titers minus twice the blank on a 5 mL fraction (indicated in the first three to four 5-mL fractions off the column), and
W = Weight of commercial monochloroacetic acid used, in grams.

$$\text{Percent monochloroacetic acid} = \frac{9.45\ A}{W}$$

where: A = The difference between the titer of the first large fraction and the appropriate blank, and
W = Weight of commercial monochloroacetic acid used, in grams.

$$\text{Percent acetic acid} = \frac{6\ G}{W}$$

where: G = The difference between the titer of the second large fraction and the appropriate blank, and
W = Weight of commercial monochloroacetic acid used, in grams.

5. PACKAGING

5.1 Unit packing. Monochloroacetic acid shall be unit packed level A, B or C and in the quantity as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 One-hundred pound quantity. A quantity of 100 (+1 or -0) pounds (lb) of monochloroacetic acid shall be unit packed in a clean, dry steel drum furnished with a bag liner. The drum shall, as a minimum, meet the requirements of a pail conforming to type II, class 11 of PPP-P-704. The bag shall conform to type I, class B, style 2 of MIL-B-117. The bag and pail shall be closed as specified in MIL-B-117 and PPP-P-704 respectively.

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5.1.1.2 Five-hundred pound quantity. A quantity of 500 (+5 or -0) lb of monochloroacetic acid shall be unit packed in the same manner as specified for the 100-lb quantity in 5.1.1.1.

5.1.2 Level B.

5.1.2.1 One-hundred pound quantity. A quantity of 100 (+1 or -0) lb of monochloroacetic acid shall be unit packed in the same manner as specified for the level A unit packing of the 100-lb quantity in 5.1.1.1, except that the drum shall conform to a fiber drum of type II, grade A of PPP-D-723, and it shall be closed as specified in PPP-D-723.

5.1.2.2 Five-hundred pound quantity. A quantity of 500 (+5 or -0) lb of monochloroacetic acid shall be unit packed in the same manner as specified for the 100-lb quantity in 5.1.2.1.

5.1.3 Level C. A quantity of 100 or 500 lb of monochloroacetic acid shall be unit packed in a manner to assure maintenance of its specified purity and quantity, and to assure compliance with DOT and any other applicable regulations, and to assure carrier acceptance and safe delivery to initial destination. Containers shall be in conformance with Uniform Freight Classification Rules, National Motor Freight Classification Rules and the rules applicable to any other intended mode of transportation.

5.2 Packing. Monochloroacetic acid, unit packed as specified in 5.1, shall require no further protection for shipment other than unitization.

5.2.1 Unitization. The 100-lb quantity, unit packed level A, shall be palletized in accordance with requirements for load type III of MIL-STD-147. The 100-lb quantity, unit packed level B, shall be palletized in accordance with requirements for load type IV of MIL-STD-147. Pallets for level A shipments shall conform to type IV, grade B of NN-P-71, while those for level B shall conform to grade A of NN-P-71. Level C shipments shall be unitized in a manner to assure carrier acceptance and safe delivery to destination, and safe, stable stackability in storage for a minimum duration of six months.

5.3 Marking. Level A and B containers and pallet loads shall be marked in accordance with MIL-STD-129. Level C containers and unitized loads shall be marked to assure identification of contents and safe prompt delivery to destination. All containers shall be marked in accordance with DOT and any other applicable regulations, and shall show the lot or batch number and date of manufacture of the monochloroacetic acid. In addition, each container shall be durably and legibly marked with the following precautionary marking:

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HAZARDS

Irritant, Eye and Skin
Toxic by Ingestion

WARNING! CAUSES IRRITATION
HARMFUL IF SWALLOWED

Avoid contact with eyes, skin, or clothing. Wash thoroughly after handling.

FIRST AID: In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Call a physician. Flush skin with water. (Wash clothing before reuse.) If swallowed and victim is conscious, have victim drink water or milk. Do not induce vomiting.

6. NOTES

6.1 Intended use. Monochloroacetic acid is intended for use in the M2A1 chemical base laboratory.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification,
(b) Level of unit packing required (see 5.1), and
(c) Unit quantity required (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.5 Sampling and testing precautions. This specification covers inspection of chemical material (monochloroacetic acid, irritant, toxic by ingestion) which is potentially hazardous to personnel. All applicable safety rules, regulations, and procedures must be followed in the handling and processing of these materials.

6.6 The Hyflow super-cel may be obtained from Johns-Manville, Artillery Row, London.

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6.7 Martin's method may be found in Symposium on partition chromatography, A. J. P. Martin, 1949, p. 11, London Biochemical Society.

Custodian:

Army - EA

Review activity:

DLA - GS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-10452C	MONOCHLOROACETIC ACID, TECHNICAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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EDITION OF 1 OCT 76 IS OBSOLETE

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